Shew Juan Kok

Senior Design & AI Engineer — AI Platform, Agentic Systems & Governance
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PROFILE

Building the AI platform, largely as its sole engineer working with IT, security and engineering teams, at a high-security semiconductor company where customer design IP sits under NDA: the gateway, retrieval, agents, evaluation and governance that let the whole engineering organisation use AI safely. I led the Claude Code rollout across an engineering team of about 100, moved company RAG from a self-built platform to AWS Bedrock, and am building an agentic project-delivery platform that turns individual AI productivity into project efficiency. I work like a forward-deployed engineer: sit with the teams, find the real problem, and ship what fits their constraints. Three years of ASIC design behind it.

EXPERIENCE

Senior Design & AI Engineer | Aion Silicon, London *Feb 2026 – Present*

Adoption and governance

- Wrote the IT security assessment and business case that got Claude approved company-wide in an IP-sensitive company, bringing engineering, IT/security and leadership to a yes.
- Led the Claude Code rollout across engineering (about 100 engineers), from pilot to daily use on every shared engineering host, connected to Microsoft 365, Jira and Freshservice through MCP, so engineers use AI on real engineering work without putting customer design IP at risk:
  - **Policy:** set by IT, and users cannot override it. Credentials stay protected, every command, edit and web fetch is logged per user, and gated upgrades with rollback keep a bad release off the fleet.
  - **Provable, not just configured:** a daily health check and a live harness that runs real Claude sessions show the controls actually work.
  - **Observability:** built a fleet dashboard for IT showing adoption and value: token use against seat cost, chat storage, blocked commands, and who is hitting limits or under-using it.
  - **Enablement:** a skills framework, taught in workshops, so engineers build their own tools instead of queueing for mine: a step-by-step procedure, about ten rules at most, and anything deterministic moved into tested code.
- Designed and delivered AWS Bedrock cost governance, so AI spend is predictable and attributed: tiered spending buckets, each capped automatically by IAM deny so one never blocks another, across 176 tagged inference profiles in 3 EU regions.

Knowledge and retrieval

- Moved company RAG from a self-built platform (LangGraph, Temporal, Weaviate, vLLM) to AWS Bedrock Knowledge Bases, now used by about 100 engineers through Open WebUI: one company-wide base, plus a base per customer project visible only to that team via IAM Identity Center and Entra ID.
- Made retrieval quality measurable (Recall@k, MRR, nDCG, faithfulness, LLM-as-judge), which overturned assumptions: replaced a reranker that was cutting nDCG by 39% with a generation model, caught 18% of the corpus silently converted to empty text, and lifted a failing sanity set from 1/10 to 8/10.
- Building doc2md, an ingestion quality gate that makes legacy documents usable by AI at lower token cost, verifying nothing is lost and generating metadata for a linked company knowledge graph.

Agentic project delivery (lead project)

- Designing and building an agentic project-delivery platform to keep complex engineering projects on time and observable. Each project is a graph of owned tasks with typed inputs and outputs, so work hands over cleanly between domains and every result traces back to the customer requirement.
- Reviewer agents do the checking and record-keeping nobody wants to do, recording a verdict per requirement in one ledger while people own intent and sign-off. On the platform's own spec they caught two requirements falsely marked done.
- Inputs are pinned by content fingerprint, so a changed file flags only the tasks that used it and errors surface before sign-off. The platform already runs its own development as about 1,600 tasks on a live status map.

Platform and infrastructure

- Designed the company AI platform's foundations around LiteLLM as a single gateway, so any team reaches any model, on Bedrock or self-hosted, through one API. Temporal workflows to orchestrate AI calls; PostgreSQL, Weaviate, Neo4j and MinIO for storage.

- Self-host models on AWS EC2 (NVIDIA L40S) with vLLM as the inference engine, serving generation and embedding models behind the gateway.
- Built the model server to be run by a team, not just me: rootless services under a shared service account, validated deploys, a watchdog, staged upgrades with rollback, and per-model GPU monitoring.
- Evaluated Bedrock against self-hosting and recommended the hybrid, avoiding paying for idle GPUs on spiky internal traffic. A feasibility study on RL fine-tuning (GRPO) for RTL review avoided building an RL platform that would not pay back.
- Built compute and licence analytics over a 3.3M-job Slurm archive and 1,300+ FlexLM features; management uses it to forecast licence use per project and cut memory waste.

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- Re-engineered a RISC-V vector core so it could drive an external VPU accelerator while keeping architectural state correct: dispatch, hazard detection and synchronised retirement.
- Built Python, Tcl and Bash automation across synthesis, lint and DFT loops to shorten design-cycle turnaround.

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- Led front-end design from architecture specification to SystemVerilog RTL across several IP and SoC projects, verified to specification with constrained-random methods and SVA.
- Set up a company-wide lint methodology (VC SpyGlass, Questa Static Formal) that runs as one command or a CI job, so every block across a project is checked against one shared rule deck; adopted by all teams.
- Owned DFT for an ASIL-B automotive SoC, debugging ATPG coverage holes in TestMAX to push test coverage above 90% and meet automotive safety requirements.
- Ran early PPA exploration in RTL Architect to surface power, performance and area risk before back-end handoff, and architected integration of third-party and in-house IP.

SELECTED PROJECTS

- **Custodex**: keeps documentation true to the code without manual audits, by detecting drift and routing each case to an LLM to fix, invalidate or escalate, with an auditable record. github.com/JuanSync7/custodex
- **experience-vault**: carries lessons from one project to the next. A nightly pipeline turns my Claude sessions and design docs into an Obsidian knowledge graph, redacting secrets and names before each push, and packages the rules I have learned as skills my other projects' agents load.
- **project-keel**: an open-source template that makes new repos agent-ready from day one, with structure and documentation gates in CI; the base of my lead project. github.com/JuanSync7/project-keel

EDUCATION

MEng Electrical and Electronic Engineering, First-Class Honours | University of Southampton *2017 – 2023*

Malaysia campus: foundation year 2017–2018, then 2020–2021. UK campus 2021–2023. National service in Singapore, Jul 2018 – Jul 2020.

SKILLS

AI and LLM: Claude Code (fleet deployment, managed policy, hooks), MCP, LangGraph, LangChain, RAG (hybrid, graph, agentic), evaluation and LLM-as-judge, OpenTelemetry

Cloud and platform: AWS (Bedrock, Knowledge Bases, EC2, IAM Identity Center, Budgets), Entra ID, vLLM, LiteLLM, FastAPI, Temporal, PostgreSQL, Weaviate, Neo4j, MinIO, Redis, Docker, Podman, Rocky/RHEL Linux, GitHub Actions, GitLab CI, Slurm

Languages: Python (primary), SystemVerilog, C/C++, Bash, Tcl

Hardware and EDA: RTL design and verification, RISC-V RVV, AMBA, DFT; Synopsys TestMAX, Design Compiler, VC SpyGlass, VCS, Verdi

Spoken: English, Mandarin Chinese, Malay (all native or bilingual)

Right to work: UK Skilled Worker visa; sponsorship transfer required